



Universität
Zürich^{UZH}

Resolving the Remaining Cases of Carlitz-Extensiveness over Finite Fields using SageMath

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Florian Hellwig & Michael Schaller

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University of Zurich

Main goal of this talk

The question

For which finite field extensions $\mathbb{F}_{q^n}/\mathbb{F}_q$ does there exist, for every $z \in \mathbb{F}_{q^n}$, an element $x \in \mathbb{F}_{q^n}$ that is both **primitive** and a **γ_z -generator**?

- Open since Hsu & Nan (2011); narrowed by Hachenberger (2016).
- **Six pairs** (q, n) remained unresolved.

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What we show

Using SageMath, we settle the six remaining cases. The complete classification:

$$\mathbb{F}_{q^n}/\mathbb{F}_q \text{ is Carlitz-extensive} \iff (q, n) \neq (2, 2).$$

The Primitive Normal Basis Theorem

The Primitive Normal Basis Theorem (PNBT)

Theorem (Lenstra & Schoof, 1987)

For every finite field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ there exists an element $\alpha \in \mathbb{F}_{q^n}$ which is simultaneously primitive and normal over $\mathbb{F}_q$.

Such an α generates $\mathbb{F}_{q^n}^\times$ multiplicatively and $\mathbb{F}_{q^n}$ additively (via its Frobenius basis).

Lenstra–Schoof

Hsu & Nan

Hachenberger

Gruber

this work

The Hsu–Nan Generalisation (2011)

The endomorphism γ_z

Hsu & Nan (2011) generalise the Primitive Normal Basis Theorem by replacing the Frobenius automorphism σ with a wider family of $\mathbb{F}_q$ -linear maps.

Definition

For $z \in \mathbb{F}_{q^n}$, define

$$\gamma_z: \mathbb{F}_{q^n} \longrightarrow \mathbb{F}_{q^n}, \quad \gamma_z(x) = z \cdot x + x^q.$$

- γ_z is an $\mathbb{F}_q$ -linear endomorphism of $\mathbb{F}_{q^n}$.
- **Special case $z = 0$:** $\gamma_0(x) = x^q = \sigma(x)$ is the Frobenius automorphism.

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Lemma (Minimal polynomial, Hsu–Nan)

If $z \in \mathbb{F}_{q^n}$ has minimal polynomial $f_z(x) \in \mathbb{F}_q[x]$ of degree $d \mid n$, then

$$m_{\gamma_z}(x) = f(x)^{n/d} - 1.$$

Definition (γ_z -generator)

An element $\alpha \in \mathbb{F}_{q^n}$ is a γ_z -**generator** if

$$\{\alpha, \gamma_z(\alpha), \gamma_z^2(\alpha), \dots, \gamma_z^{n-1}(\alpha)\} \quad \text{is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

- A γ_z -generator **always exists** but does **not need to be primitive**.

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Definition (Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is **Carlitz-extensive** if for every $z \in \mathbb{F}_{q^n}$ there exists a primitive γ_z -generator.

Hsu & Nan: open cases

Hsu & Nan worked with character-sum estimates and solved all but 63 pairs (q, n) pairs. These 63 pairs were too small for asymptotic methods to apply.

Known exception

$(q, n) = (2, 2)$, i.e. $\mathbb{F}_4/\mathbb{F}_2$, is *not* Carlitz-extensive. Is it the only one?

Hsu & Nan, J. Number Theory 131 (2011), 146–157.

(Subsequent work reduced the 63 open cases to just six — as we will see.)



The Hachenberger Generalisation (2016)

Hachenberger: cyclic vector spaces

Hachenberger (2016) embeds the problem in a wider framework: replace γ_z by *any* $\mathbb{F}_q$ -linear endomorphism τ of $\mathbb{F}_{q^n}$ for which a τ -generator exists (analogue to γ_z -generator). Such a τ is then called *cyclic*.

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Definition (Extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is *extensive* if for every cyclic $\mathbb{F}_q$ -endomorphism τ of $\mathbb{F}_{q^n}$ there exists a primitive τ -generator $\alpha \in \mathbb{F}_{q^n}$, i.e. α generates both $\mathbb{F}_{q^n}^\times$ and $\mathbb{F}_{q^n}$ as an $\mathbb{F}_q[\tau]$ -module.

Specialisation: $\tau = \gamma_z, z \in \mathbb{F}_{q^n} \Rightarrow$ primitive γ_z -generator $\rightarrow$ Hsu–Nan

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Specialisation: $\tau = \gamma_z, z \in \mathbb{F}_{q^n} \Rightarrow$ primitive γ_z -generator $\rightarrow$ Hsu–Nan

Progress on the 63 open cases:

- Hachenberger resolved 41 of them.
- Master's thesis of Thomas Gruber reduced the list by another 16 pairs.

Hachenberger, Australasian J. Combin. 64 (2016), 289–326.



The remaining 6 cases

After Hachenberger and Gruber, only **six pairs** (q, n) remained unresolved:

q	n
2	18, 20, 24
4	10, 12
8	8

- **Too large** for naive brute force.
- **Too small** for asymptotic / structural arguments.

Goal of this work: settle each case by an explicit computation in SageMath.



Our Approach: Reductions & Algorithm

Strategy at a glance

We reduce the search using:

1. **Cyclotomic cosets**: one z per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial f_z of z .

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Strategy at a glance

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1. **Cyclotomic cosets**: one z per orbit (corresponding to coset) suffices (outer loop), simplifies the calculation of the minimal polynomial f_z of z .
2. **Projective space**: one x per projective line suffices (inner loop), restrict the search for a primitive γ_z -generator to one representative per projective primitive line.
3. **Minimal-polynomial factorisation**: fast algebraic test for the γ_z -generator property, more efficient than e.g. the basis check.

Each tool decreases the number of computations needed, making the algorithm feasible.

The algorithm in one slide

Input. A pair (q, n) . **Output.** True / False for Carlitz-extensiveness.

Preparation.

1. Fix a primitive element $g \in \mathbb{F}_{q^n}^\times$; set $N = q^n - 1$, $Q = N/(q - 1)$.

2. Precompute representatives of the primitive projective points:

$$\mathcal{R} = \{g^k : 1 \leq k < Q, \gcd(k, Q) = 1\}.$$

3. Handle $z = 0$ separately (Frobenius case): search $\mathcal{R}$ for a primitive γ_0 -generator.

Main loop. For each q -cyclotomic coset of $\mathbb{Z}/N\mathbb{Z}$:

1. Pick representative $z = g^e$ of Frobenius orbit.
2. Compute the minimal polynomial f_z of z via its cyclotomic coset, then factor $m_{\gamma_z} = f_z^{n/\deg f_z} - 1 = \prod_i p_i^{e_i}$.
3. Build the matrix A of γ_z , then $H_i = (m_{\gamma_z}/p_i)(A)$ for each i .
4. Scan $\mathcal{R}$: if $\text{Vec}(g^k) \cdot H_i \neq 0$ for all i , then $\alpha = \lambda g^k$ is a primitive γ_z -generator.
5. If $\mathcal{R}$ is exhausted with no hit: return False.

If every coset succeeds: return True.

Results

Main result

Theorem (Hellwig & Schaller, 2026)

Each of the six remaining pairs

$$(q, n) \in \{(2, 18), (2, 20), (2, 24), (4, 10), (4, 12), (8, 8)\}$$

is Carlitz-extensive.

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is *Carlitz-extensive*.

Combined classification

A finite-field extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Carlitz-extensive *if and only if* $(q, n) \neq (2, 2)$.

Equivalently, $\mathbb{F}_4/\mathbb{F}_2$ is the *unique* non-Carlitz-extensive extension among all finite extensions of finite fields.

Two independent searches

- **Original algorithm** (projective-point first): $\approx 18\text{h}$ on one core for all six cases together.
- **Alternative algorithm** (generator-first): find one γ_z -generator u , then search candidates $a(\gamma_z)(u)$ with $a \in \mathbb{F}_q[x]$ monic, $\deg a < n$, $\gcd(a, m_{\gamma_z}) = 1$. $\approx 21\text{h}$ on one core.

Both produce identical results on all six cases.

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Both produce identical results on all six cases.

Repository

Code, pseudocode, benchmarks, and additional material:

`github.com/florian-hellwig/Carlitz-extensiveness`

Outlook: Generalisations and Conjectures

$\mathbb{F}_q[x]$ -Drinfeld extensiveness: an intermediate framework

Let $\tau: x \mapsto x^q$ be the Frobenius on $\mathbb{F}_{q^n}$.

Definition ($\mathbb{F}_q[x]$ -Drinfeld endomorphism of rank r)

For $t, c_1, \dots, c_r \in \mathbb{F}_{q^n}$ with $c_r \neq 0$,

$$\varphi = t \cdot \tau^0 + c_1 \tau + \dots + c_r \tau^r \in \text{End}_{\mathbb{F}_q}(\mathbb{F}_{q^n}), \quad \varphi(v) = tv + c_1 v^q + \dots + c_r v^{q^r}.$$

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Definition ($\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is $\mathbb{F}_q[x]$ -**Drinfeld-extensive of rank $\leq r$** if for every *cyclic* $\mathbb{F}_q[x]$ -Drinfeld endomorphism φ of rank $\leq r$, there exists a primitive φ -generator $\alpha \in \mathbb{F}_{q^n}$.

- Any $\mathbb{F}_q[x]$ -Drinfeld endomorphism of rank $r = 1$ is *cyclic*. Not the case for $r \geq 2$.
- $r \leq 1$ with $c_1 = 1 \implies \varphi = \gamma_t$ & *Carlitz-extensiveness*
- $r \geq n - 1$: Any $\mathbb{F}_q$ -linear map is $\mathbb{F}_q[x]$ -Drinfeld polynomial $\rightarrow$ Hachenberger-*extensiveness*

$\mathbb{F}_q[x]$ -Drinfeld-Extensiveness: Classification

Hierarchy

Extensive $\implies \mathbb{F}_q[x]$ -Drinfeld-extensive of rank $\leq r$ for every $r \geq 1$.

Known non-extensive pairs:

(q, n)	$\mathbb{F}_q[x]$ -Drinfeld status
$(2, 2)$	Not Carlitz-ext. $\Rightarrow$ not $\mathbb{F}_q[x]$ -Drinfeld-ext. rank ≤ 1
$(3, 2), (5, 2), (2, 4)$	Carlitz-ext., not $\mathbb{F}_q[x]$ -Drinfeld-ext. rank ≤ 1
$(2, 6)$	$\mathbb{F}_q[x]$ -Drinfeld-ext. rank ≤ 2 , not rank ≤ 3

Extensiveness unknown:

$(2, 8), (2, 10), (2, 12), (2, 14),$
 $(2, 15), (2, 16), (2, 18), (2, 20),$
 $(2, 24), (3, 8), (3, 10), (3, 12),$
 $(4, 6), (4, 9), (4, 10), (4, 12),$
 $(5, 4), (7, 6), (8, 8).$

$\mathbb{F}_q[x]$ -Drinfeld-Extensiveness: Classification

Hierarchy

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Extensiveness unknown:

$(2, 8), (2, 10), (2, 12), (2, 14), (2, 15), (2, 16), (2, 18), (2, 20), (2, 24), (3, 8), (3, 10), (3, 12), (4, 6), (4, 9), (4, 10), (4, 12), (5, 4), (7, 6), (8, 8).$

Result (Hellwig & Schaller, 2026)

We prove $(5, 4)$ is **extensive**, hence $\mathbb{F}_q[x]$ -Drinfeld-extensive of every rank.

All remaining cases are **conjectured extensive**.

Simultaneous extensiveness: matching every subfield level

New question: Find one primitive $\alpha \in \mathbb{F}_{q^n}$ that is a generator simultaneously at every intermediate field $\mathbb{F}_q \subseteq \mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$, $d \mid n$.

Note that for $d = n$, any $\alpha \neq 0$ is trivially a generator.

- **Simultaneous Carlitz-extensiveness of $\mathbb{F}_{q^n}/\mathbb{F}_q$:** for any $z \in \mathbb{F}_{q^n}$ for which a simultaneous $\gamma_z^{(d)}$ -generator over $\mathbb{F}_{q^d}$ exists for all $d \mid n$, $d < n$, where $\gamma_z^{(d)}(v) = zv + v^{q^d}$, there exists a *primitive* simultaneous $\gamma_z^{(d)}$ -generator.

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- **Simultaneous $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$:** for any $t, c_1, \dots, c_r \in \mathbb{F}_{q^n}$ for which a simultaneous $\varphi^{(d)}$ -generator over $\mathbb{F}_{q^d}$ exists for all $d \mid n$, $d < n$, where $\varphi^{(d)}(v) = tv + c_1 v^{q^d} + \dots + c_r v^{q^{rd}}$, there exists a *primitive* simultaneous $\varphi^{(d)}$ -generator.

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- **Simultaneous extensiveness:** for any family $(\tau_d)_{d \mid n, d < n}$ of $\mathbb{F}_{q^d}$ -linear endomorphisms of $\mathbb{F}_{q^n}$ admitting a simultaneous generator, there exists a *primitive* simultaneous generator.

Equivalently: Simultaneous $\mathbb{F}_q[x]$ -Drinfeld-extens. with $r \geq n - 1$ and coeffs not fixed.

Fixing $\varphi^{(d)}(v) = \sigma^d(v) = v^{q^d}$ recovers the **Morgan–Mullen conjecture**.

Three conjectures on simultaneous extensiveness

Conjecture (Simultaneous Carlitz-extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously Carlitz-extensive for *all but finitely many* pairs (q, n) .

Conjecture (Simultaneous $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$)

For each fixed $r \geq 1$, $\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously $\mathbb{F}_q[x]$ -Drinfeld-extensive of rank $\leq r$ for *all but finitely many* pairs (q, n) .

Conjecture (Simultaneous extensiveness)

$\mathbb{F}_{q^n}/\mathbb{F}_q$ is simultaneously extensive for *all but finitely many* pairs (q, n) .

Simultaneous extensiveness $\implies$ Simultaneous $\mathbb{F}_q[x]$ -Drinfeld-extensiveness of rank $\leq r$
 $\implies$ Simultaneous Carlitz-extensiveness

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- [4] THE SAGE DEVELOPERS, *SageMath, the Sage Mathematics Software System*.
<https://www.sagemath.org>

Thank you!

Questions?

Primitive and normal elements

Let $\mathbb{F}_{q^n}/\mathbb{F}_q$ be a finite field extension of degree n .

Primitive element ($\rightarrow$ multiplicative structure)

$\alpha \in \mathbb{F}_{q^n}$ is **primitive** if $\mathbb{F}_{q^n}^\times = \langle \alpha \rangle$. Equivalently, $\text{ord}(\alpha) = q^n - 1$.

Normal element ($\rightarrow$ additive structure)

$\alpha \in \mathbb{F}_{q^n}$ is **normal** over $\mathbb{F}_q$ if the iterates of the Frobenius automorphism form an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^n}$:

$$\{\alpha, \alpha^q, \alpha^{q^2}, \dots, \alpha^{q^{n-1}}\} \text{ is an } \mathbb{F}_q\text{-basis of } \mathbb{F}_{q^n}.$$

Counterexample: $\mathbb{F}_4/\mathbb{F}_2$ is not Carlitz-extensive

Setup. $\mathbb{F}_{q^n} = \mathbb{F}_{2^2} = \mathbb{F}_4 = \mathbb{F}_2[x]/(x^2 + x + 1)$, $\mathbb{F}_q = \mathbb{F}_2$. α root of $x^2 + x + 1$, so $\alpha^2 = \alpha + 1$.

- $|\mathbb{F}_4^\times| = 3$ is prime $\implies$ primitive elements are $\{\alpha, \alpha + 1\}$
- x is a γ_z -generator $\iff \{x, \gamma_z(x)\}$ is an $\mathbb{F}_2$ -basis, where $\gamma_z(x) = z \cdot x + x^2$

z	$\{\alpha, \gamma_z(\alpha)\}$	$\{\alpha + 1, \gamma_z(\alpha + 1)\}$
0	$\{\alpha, \alpha + 1\}$ ✓	$\{\alpha + 1, \alpha\}$ ✓
1	$\{\alpha, 1\}$ ✓	$\{\alpha + 1, 1\}$ ✓
α	$\{\alpha, 0\}$ ✗	$\{\alpha + 1, \alpha + 1\}$ ✗
$\alpha + 1$	$\{\alpha, \alpha\}$ ✗	$\{\alpha + 1, 0\}$ ✗

Conclusion: For $z \in \{\alpha, \alpha + 1\}$, only 1 but no primitive element is a γ_z -generator

Cyclotomic cosets and Frobenius compatibility

Cyclotomic coset (modulo $q^n - 1$)

$$C_k = \{k, kq, kq^2, \dots\} \pmod{q^n - 1}, \quad |C_k| \mid n.$$

For a primitive $g \in \mathbb{F}_{q^n}$, exponentiation turns C_k into the **Frobenius orbit** of g^k , and yields the minimal polynomial $f_{g^k} = \prod_{j \in C_k} (x - g^j)$ easily.

Lemma (Conjugation lemma)

For every $z \in \mathbb{F}_{q^n}$ and $i \in \mathbb{Z}$, $\gamma_{\sigma^i(z)} = \sigma^i \circ \gamma_z \circ \sigma^{n-i}$.

- If α is a primitive γ_z -generator, then $\sigma^i(\alpha)$ is a primitive $\gamma_{\sigma^i(z)}$ -generator.
- **Reduction:** test only **one z per Frobenius orbit**.

Projective space and $\mathbb{F}_q^\times$ -invariance

Set $N = q^n - 1$ and $Q = N/(q - 1)$. The projective $(n-1)$ -space:

$$\mathbb{P}_{\mathbb{F}_q}^{n-1} = \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times \quad (\text{cyclic of order } Q).$$

A projective point $[g^k]$ is **primitive** if it generates this quotient, i.e. $\gcd(k, Q) = 1$.

Lemma ($\mathbb{F}_q^\times$ -invariance)

Let $z, x \in \mathbb{F}_{q^n}$ and $\lambda \in \mathbb{F}_q^\times$. Then

$$x \text{ is a } \gamma_z\text{-generator} \iff \lambda x \text{ is a } \gamma_z\text{-generator}.$$

Reduction: the γ_z -generator property depends only on the projective point $[x]$. Once a primitive projective point that is a γ_z -generator is found, some scalar multiple λx is **automatically a primitive γ_z -generator**.

A prerequisite: from individual to simultaneous cyclicity

The simultaneous extensiveness conjectures require a simultaneous (non-primitive) generator as hypothesis. When is this guaranteed?

Conjecture (Individual $\Rightarrow$ simultaneous cyclicity)

Let $(\tau_d)_{d|n, d < n}$ be a family of individually cyclic $\mathbb{F}_{q^d}$ -linear endomorphisms of $\mathbb{F}_{q^n}$. Then a simultaneous generator exists, i.e., there is $\alpha \in \mathbb{F}_{q^n}$ with

$$\mathbb{F}_{q^d}[\tau_d] \cdot \alpha = \mathbb{F}_{q^n} \quad \text{for all } d \mid n, d < n.$$

- $\tau_d = \sigma^d$: Existence of completely normal elements proven by [Blessenohl–Johnsen \(1986\)](#).
- $\mathbb{F}_q[x]$ -Drinfeld special case $\tau_d = \varphi^{(d)}$: Fix $z, c_1, \dots, c_r \in \mathbb{F}_{q^n}$.

$$\forall d \mid n, d < n, \exists \text{ a } \varphi^{(d)}\text{-generator } \alpha \in \mathbb{F}_{q^n} \stackrel{?}{\implies} \exists \text{ a } \left(\varphi^{(d)} \right)_{d|n, d < n}\text{-generator } \alpha \in \mathbb{F}_{q^n}$$

- [Carlitz-special case](#) with $\varphi^{(d)} = \gamma_z^{(d)}$.